



Software Release Notes for the Packaging and Deployment Solutions

Release 1.4.3

Date: December 2022

Affirmed Networks, Inc.
35 Nagog Park
Acton, MA 01720
Tel: +1-978-268-0800
www.affirmednetworks.com

A Microsoft Company

Copyright© 2022 Affirmed Networks, Inc. All Rights Reserved.

AFFIRMED & DESIGN®, ACUITAS SERVICE MANAGEMENT SYSTEM®, AFFIRMED NETWORKS MOBILE CONTENT CLOUD™, AFFIRMED MOBILE CONTENT CLOUD™, UnityCloud™, UnityCloud PaaS™, UnityCloud Platform™, UnityCloud Operations™, UnityCloud Ops™, Operations and Policy Manager (OPM)™ and other trademarks and designs are the registered or unregistered trademarks of Affirmed Networks, Inc. and its subsidiaries in the United States and in foreign countries. All other trademarks are the property of their respective owners. The Affirmed Networks, Inc. trademarks may not be used in connection with any product or service that is not Affirmed Networks' in any manner that is likely to cause confusion among customers or in any manner that disparages or discredits Affirmed Networks, Inc.

This document contains information that is the property of Affirmed Networks, Inc. This document may not be copied, reproduced, reduced to any electronic medium or machine readable form, or otherwise duplicated, and the information herein may not be used, disseminated or otherwise disclosed, except with the prior written consent of Affirmed Networks, Inc.

Contents

Release Abstract	4
New Features and Enhancements	5
Known Anomalies	8
Major Anomalies	8
Problems Fixed	8
Install PADS Software	9
Prerequisites	9
Installing the PADS Software	9
LIB-CODE	9
INFRA-DEV	9
SVC-DEV	10
Upgrade PADS Software	11
Prerequisites	11
Steps to Upgrade	11
LIB-CODE	11
INFRA-DEV	12
SVC-DEV	12

Release Abstract

This document contains the software release notes for the Affirmed Networks Packaging and Deployment Solutions, Release 1.4.3.

New Features and Enhancements

This release contains the following new features and enhancements in the ACR to OPM Image Copy, CNF Delete, CNF Deploy, and Finalize K8s Cluster Pipelines and their corresponding repositories (INFRA-DEV, LIB-CODE, or SVC-DEV) and files.

Feature	Description
ACR to OPM Image Copy Pipeline (INFRA-DEV) acr-to-opm-image-copy-macro-pipeline.yml	Added the following variables: ▪ name: Name of the Pipeline when it is run. Includes parameter zoneName. ▪ appendCommitMessageToRunName: Append the commit message to the build number. The default is true. ▪ zoneName: Parameter to specify zone name. ▪ azureRegion: Parameter for the Azure region the Pipeline runs on. ▪ versionSpecificSourceFolder: Directory path to version specific file. ▪ versionspecific: Name of the version specific file.
CNF Delete Pipeline (SVC-DEV) cnf-operations-delete-macro-pipeline.yml	The pipeline calls the OPM's NB API to delete all federations in the given cluster. The pipeline polls for all federations to be deleted until the timeout is reached. If all of the federations are deleted, the pipeline continues the rook-ceph cleanup process. If all federations are not deleted and some pods are stuck in the Terminating state, then the pipeline exits out after timeout. A manual intervention is then needed to understand why the pods are not deleted and must be cleaned up using kubectl delete pod commands, such as: kc delete pods -n fed-upf --force --all. See Known Anomalies. Added the following variables: ▪ pool.name: Name of the Agent Pool. ▪ resources.repositories.repository: Name of repository used across the pipeline. ▪ resources.repositories.type: Type of repository. ▪ resources.repositories.name: Name of the actual ADO repository. ▪ resources.repositories.ref: Branch of the ADO repository ▪ resources.containers.container: Container reference name used across the pipeline. ▪ resources.containers.image: Image name with reference to ACR. ▪ zoneName: Name of zone used to name the pipeline run. ▪ cleanup: Cleanup process after CNF Delete. Default is true. ▪ stages.template: Path to micro pipeline with the corresponding ADO repository.

(LIB-CODE) cnf-operations-delete.json	Added the following variables: ▪ nfType: Type of CNF being deployed. (SMF/UPF) ▪ kubeconfig.reference.keyVault.id: Name of the Azure Key-Vault ▪ kubeconfig.reference.keyVault.secretName: Name of the secret in Azure Key-Vault with value as kubeconfig of the K8s Cluster. ▪ opm.ipAddress: IP address of OPM. ▪ opm.username: Username used to authenticate to OPM application. ▪ opm.keyVault.opmPasswordSecretName: Secret name in Azure Key-Vault of opm.username. ▪ snapshotFolder: File path in LIB-CODE to snapshot folder. ▪ snapshotFile: Name of Templated Snapshot file in LIB-CODE's snapshotFolder. ▪ snapshotVariableValuesFile: Name of snapshot variables/values file in snapshotFolder. ▪ chartRepo: Name of Chart Repository created on OPM. This will be used by OPM to do Bulk Install using your snapshot. ▪ dzName: Name of Deployment Zone to which K8s Cluster is attached to. ▪ clusterId: ID of the attached K8s Cluster from OPM. User can get this from FinalizeK8sCluster Pipeline Attach K8s Cluster stage. User can get this from the FinalizeK8sCluster Pipeline Attach K8s Cluster stage. ▪ cephDiskType: Takes values either vdc for SMF or vdb for UPF.
CNF Deploy Pipeline (SVC-DEV) cnf-operations-deploy-macro-pipeline.yml	The wording in micro and macro pipeline file and folder names are modified from helm to cnf. Added, updated, and removed the following variables: Added ▪ name: Name of the Pipeline when it is run. Includes parameter zoneName. ▪ appendCommitMessageToRunName: Append the commit message to the build number. The default is true. ▪ fedDeployTimeout: Polling interval for calling NB API to poll all federations status. Value in seconds. Recommended value 10. ▪ zoneName: Parameter to specify zone name. ▪ workerCount: Parameter for number of workers. ▪ azureRegion: Parameter for the Azure region the Pipeline runs on. ▪ versionSpecificSourceFolder: Directory path to version specific file. ▪ versionspecific: Name of the version specific file. Updated ▪ cnfDeployTimeout: Name is modified from original helmDeployTimeout. Removed ▪ deployTimeout ▪ perFedMaxPollTimeout

(LIB-CODE) cnf-operations-deploy.json	Added and removed the following variables: Added ▪ opm.ipAddress: IP address of OPM. ▪ opm.username: Username used to authenticate to OPM application. ▪ opm.keyVault.opmPasswordSecretName: Secret name in Azure Key-Vault of opm.username. ▪ snapshotFolder: File path in LIB-CODE to snapshot folder. ▪ snapshotFile: Name of Templated Snapshot file in LIB-CODE's snapshotFolder. ▪ snapshotVariableValuesFile: Name of snapshot variables/values file in snapshotFolder. ▪ chartRepo: Name of Chart Repository created on OPM. This will be used by OPM to do Bulk Install using your snapshot. ▪ dzName: Name of Deployment Zone to which K8s Cluster is attached to. ▪ clusterId: ID of the attached K8s Cluster from OPM. User can get this from FinalizeK8sCluster Pipeline Attach K8s Cluster stage. Removed ▪ acrInfo.name ▪ acrInfo.storageFormat ▪ prdLoction ▪ helmValuesLocation ▪ dockerRepo
(LIB-CODE) code/cnf_operations/snapshots/snapshot_specific_values.json	This file contains the values to the variables in the snapshot file used by the CNF Operation Deploy Pipeline. The following is a sample for assigning value variables: \$ml_peer, \$peer_as: <pre> { "\$ml_peer": "10.161.30.8", "\$peer_as": "64500" } </pre>
Finalize K8s Cluster Pipeline	Validation for HA VIP is added to the FinalizeK8sCluster Pipeline after the Master 2 node joins the K8s Cluster as part of the FINALIZE K8s Cluster task in the Finalize K8s Cluster stage. If validation fails, the pipeline exits with an error code and can be viewed in the pipeline log.

(INFRA-DEV) finalizeK8sClustermacro- pipeline.yml	Added the following variables: ▪ name: Name of the Pipeline when it is run. Includes parameter zoneName. ▪ appendCommitMessageToRunName: Append the commit message to the build number. The default is true. ▪ zoneName: Parameter to specify zone name. ▪ workerCount: Parameter for number of workers. ▪ azureRegion: Parameter for the Azure region the Pipeline runs on. ▪ forceLocalDNS: Parameter to fix local dns issue with VM images. ▪ versionSpecificSourceFolder: Directory path to version specific file. ▪ versionspecific: Name of the version specific file. ▪ forcelocalDnsPods: True/False parameter if fix for force local DNS pods must be applied after K8s Cluster creation.
(LIB-CODE) finalizeK8sCluster.json	Added the following variables: ▪ opm.username: OPM GUI username. ▪ opm.opmPasswordSecretName: Azure Key-Vault secret name for OPM GUI user's password. ▪ opm.dzName: Deployment Zone name on OPM to which the create K8s Cluster needs to be attached.
PADS BOM	The PADS BOM YAML file is renamed to bom-pads-<release-build>.yaml format.

Known Anomalies

This section lists the known anomalies in this release.

Major Anomalies

This release contains the following known anomaly.

Anomaly	Description
The CNF Delete pipeline is unsuccessful.	If all federations are not deleted, some pods remain in the Terminating state and the pipeline exits after timeout. Manual intervention is required to understand why the pods are not deleted and must be cleaned up using kubectl delete pod commands, such as: <pre>kc delete pods -n fed-upf --force --all</pre>

Problems Fixed

This release contains only new features and enhancements.

Install PADS Software

Prerequisites

Verify that the following prerequisite tasks are complete prior to installing the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. Extract the `pipeline_artifacts.tar.gz` file inside `pads_pipeline-<version>-<build_number>/.`
 - The results should show the 3 directories LIB-CODE/, INFRA-DEV/ and SVC-DEV/ extracted.
- Confirm that the following repositories exist on the ADO project: LIB-CODE, INFRA-DEV, and SVC-DEV.
- Confirm that Service Connections to the following exist on the ADO project: Azure Resource Group and Azure Container Registry.

Installing the PADS Software

Complete the following steps in each ADO repository.

LIB-CODE

1. Copy the contents of the *LIB-CODE/* directory extracted from *pipeline_artifacts.tar.gz* into the LIB-CODE ADO repository.
2. Update the *code/acr_opm_image_copy/acrToOpmCopy.json* file with appropriate values.
3. Update the *code/finalize_cluster/finalizeK8sCluster.json* file with appropriate values.
4. Update the following files with the appropriate files.
 - *code/cnf_operations/cnf-operations-deploy.json*
 - *code/cnf_operations/cnf-operations-delete.json*
5. Add the *cna_deploy_inventory* script file to a folder in LIB-CODE and refer to it in the appropriate variables.
6. Add the *snapshotFile* value and *snapshot_specific_values.json* files to LIB-CODE and refer to them in the appropriate variables.

INFRA-DEV

1. Copy the contents of the INFRA-DEV/ directory extracted from *pipeline_artifacts.tar.gz* into the INFRA-DEV ADO repository.
2. Update the values in the following macro pipeline files: *acr-to-opm-image-copy-macro-pipeline.yml* and *finalizeK8sCluster-macro-pipeline.yml*.

SVC-DEV

1. Copy the contents of the SVC-DEV/ directory extracted from *pipeline_artifacts.tar.gz* into the SVC-DEV ADO repository.
2. Update the values in the macro pipeline files, *cnf-operations-delete-macro-pipeline.yml* and *cnf-operations-deploy-macro-pipeline.yml*.

Upgrade PADS Software

Prerequisites

Verify that the following prerequisite tasks are complete prior to upgrading the PADS software:

- Extract the `pads_pipeline-<version>-<build_number>.tgz` file. Extract the `pipeline_artifacts.tar.gz` file inside `pads_pipeline-<version>-<build_number>/.`
 - The results should show the 3 directories LIB-CODE/, INFRA-DEV/ and SVC-DEV/ extracted.
- Confirm that the following repositories exist on the ADO project: LIB-CODE, INFRA-DEV, and SVC-DEV.
- Confirm that Service Connections to the following exist on the ADO project: Azure Resource Group and Azure Container Registry.

Steps to Upgrade

Complete the following steps in each ADO repository to upgrade.

LIB-CODE

1. Override all files in `pipeline/` with the contents of `LIB-CODE/pipeline/` directory extracted from `pipeline_artifacts.tar.gz`.
2. Add new variables to `code/acr_opm_image_copy/acrToOpmCopy.json`.
 - Use `LIB-CODE/code/acr_opm_image_copy/acrToOpmCopy.json` extracted from `pipeline_artifacts.tar.gz` for reference.
3. Add new variables to `code/finalize_cluster/finalizeK8sCluster.json` file.
 - Use `LIB-CODE/code/finalize_cluster/finalizeK8sCluster.json` extracted from `pipeline_artifacts.tar.gz` for reference.
4. Update below files with appropriate variables from Release Notes. Use the corresponding `LIB-CODE/` directory files for reference.
 - `code/cnf_operations/cnf-operations-deploy.json`
 - `code/cnf_operations/cnf-operations-delete.json`
5. Add the `cna_deploy_inventory` script file to a folder in LIB-CODE and refer to the script file.
6. Add the PRD files to `code/acr_opm_image_copy` and refer to the PRD files.

INFRA-DEV

Update the values in the following macro pipeline files: `acr-to-opm-image-copy-macro-pipeline.yml` and `finalizeK8sCluster-macro-pipeline.yml`.

SVC-DEV

Update the values in the macro pipeline files, `cnf-operations-delete-macro-pipeline.yml` and `cnf-operations-deploy-macro-pipeline.yml`.